

Anakin Dey

Curriculum Vitæ

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Research Interests

Algebraic and Geometric Complexity Theory, Algebraic Combinatorics, Computational Algebra

Education

Fall 2024 — **PhD Mathematics**, *The Ohio State University*

Spring 2030 Advised by Dave Anderson and Zeyu Guo

Fall 2020 — **BSLAS Mathematics**, *University of Illinois Urbana-Champaign*

Spring 2024 Minor in Computer Science

Graduation with Highest Distinction

Publications

2025 **Debordering Closure Results in Determinantal and Pfaffian Ideals**

Anakin Dey and Zeyu Guo

Innovations in Theoretical Computer Science 2026, [LIPIcs](#)

2025 **Optimizing a Model-Agnostic Measure of Graph Counterdeceptiveness via Reattachment**

Anakin Dey, Sam Ruggerio, Manav Vora, and Melkior Ornik

IEEE Cyber Awareness and Research Symposium 2025, [IEEE](#)

2024 **Classifying Primitive Solvable Permutation Groups of Rank 5 and 6**

Anakin Dey, Kolton O'Neal, Duc Van Khanh Tran, Camron Upshur, and Yong Yang

Journal of Group Theory, [Degruyter](#)

2022 **Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency**

Anakin Dey and Melkior Ornik

IEEE International Smart Cities Conference 2022, [IEEE](#)

Talks

2026 **What Is Hilbert's 17th Problem?**

OSU What Is...? Seminar

2026 **Debordering Closure Results in Determinantal and Pfaffian Ideals**

9th Workshop on Algebraic Complexity Theory

2026 **Symmetric Functions and the Affine Symmetric Group**

RTG 2026 Workshop: Affine Combinatorics and Quantum K-Theory

2026 **Debordering Closure Results in Determinantal and Pfaffian Ideals**

UIUC Theory Seminar

2026 **Debordering Closure Results in Determinantal and Pfaffian Ideals**

Purdue Theory CS/Math Seminar

- 2026 **Debordering Closure Results in Determinantal and Pfaffian Ideals**
Innovations in Theoretical Computer Science 2026
- 2025 **The Affine Symmetric Group**
OSU Affine Grassmannian Seminar
- 2025 **What Is Algebraic Complexity Theory?**
OSU What Is...? Seminar
- 2024 **Classifying Primitive Groups using Computer Algebra and Change Making**
China-US Group Theory Summit 2024
- 2023 **Classifying Primitive Groups using Computer Algebra and Change Making**
UIUC Department of Mathematics Undergraduate Friday Seminars
- 2023 **Classifying Primitive Solvable Permutation Groups of Rank 5 and 6**
China Three Gorges University Algebra and Combinatorics Seminar
- 2022 **Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency**
IEEE International Smart Cities Conference 2022

Research Experience

- Spring 2024 **Undergraduate Research Assistant**, [ICLUE](#)
Under Prof. Alexander Yong, University of Illinois Urbana-Champaign
- Summer 2023 **Undergraduate Research Assistant**, [REU in Algebra, Combinatorics, and Statistics](#)
Under Prof. Yong Yang, Texas State University
- Fall 2021 — **Undergraduate Research Assistant**, [LEADCAT Lab](#)
Fall 2023 Under Prof. Melkior Ornik, University of Illinois Urbana-Champaign

Student Organizations

- Spring 2026 – **Committee Member**, OSU [Cycle](#)
Current Ohio State University Cycle
- Spring 2025 – **Vice President**, OSU [MGSA](#)
Current Ohio State University Math Graduate Student Association
- Fall 2022 – **Founder and President**, ACM [SIGma](#)
Spring 2024 Special Interest Group in Math and Algorithms
- Fall 2021 – **Admin**, ACM [SIGPwny](#)
Spring 2024 Special Interest Group in Cybersecurity

Teaching Experience

- Fall 2025 – **Graduate Teaching Associate**, The Ohio State University
Current
 - Math 3345 — Foundations of Higher Mathematics (Spring 2026, Grader)
 - Math 1151 — Calculus I (Fall 2025, TA)

Spring 2021 – **Undergraduate Course Assistant**, University of Illinois Urbana-Champaign

Spring 2024 ○ CS 225H — Honors Data Structures
(*Spring 2024*)

○ CS 374 — Introduction to Algorithms and Models of Computation
(*Spring 2022* — *Fall 2022*, *Fall 2023*)

○ CS 173 — Discrete Structures
(*Spring 2021* — *Spring 2024*)

○ CS 475 — Models of Computation
(*Spring 2023*)

○ CS 125 — Introduction to Computer Science
(*Spring 2021*)

Organization

Fall 2026 **Algebraic Geometry Reading Group**

Spring 2026 **Cycle/MGSA/Spectra Puzzlehunt**

Spring 2026 **5th Annual OSU Cycle Conference**

Spring 2026 **Commutative Algebra Reading Group**

Spring 2025 – **MGSA Seminar**
Current

Fellowships and Awards

Fall 2026 **David Goss Technology & Academic Innovation Stimulus Fund**
The Ohio State University Department of Mathematics

Fall 2024 — **University Fellowship**
Spring 2025 The Ohio State University Graduate School

Posters

2026 **Debordering Closure Results in Determinantal and Pfaffian Ideals**
AlGeCom XXVII

2026 **Debordering Closure Results in Determinantal and Pfaffian Ideals**
OSU Computer Science and Engineering Poster Expo

2023 **Classifying Primitive Solvable Permutation Groups of Rank 5**
TXST REU Poster Session

Professional Experience

Summer 2022 **Software Engineering Intern**, CME Group

Summer 2021 **Software Engineering Intern**, SitelQ